

RBG-48244

10 CFR 50.73

August 17, 2023

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: Licensee Event Report 50-458 / 2023-001-01, Ultimate Heat Sink
Inoperable due to Boundary Valve Leakage

River Bend Station – Unit 1
NRC Docket Nos. 50-458
Renewed Facility Operating License No. NPF-47

On May 23, 2023, Licensee Event Report 50-458 / 2023-001-00, Ultimate Heat Sink Inoperable due to Boundary Valve Leakage (ML23143A276) was submitted by Entergy Nuclear Operating Inc. This letter is a supplement to the report to provide the updated results of the causal evaluation. The updated information is denoted by revision bars located in the right-hand margin. The report is submitted in accordance with 10 CFR 50.73.

This document contains no commitments.

Should you have any questions, please contact Mr. Randy Crawford, Regulatory Assurance Manager, at 225-381-4177.

Respectfully,



Phil Hansett

PH/twf

Enclosure: Licensee Event Report 50-458 / 2023-001-01, Ultimate Heat Sink Inoperable
due to Boundary Valve Leakage

cc: NRC Region IV Regional Administrator - Region IV
NRC Senior Resident Inspector - River Bend Station

Enclosure

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**Licensee Event Report 50-458 / 2023-001-01, Ultimate Heat Sink Inoperable due to
Boundary Valve Leakage**

NRC FORM 366 (03-14-2023)		U.S. NUCLEAR REGULATORY COMMISSION		APPROVED BY OMB: NO. 3150-0104		EXPIRES: 08/31/2023					
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/stat/sr1022r3/ Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA Library and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to InfoCollectionsResource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: ora_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.											
1. Facility Name River Bend Station, Unit 1				☒ 050 ☐ 052		2. Docket Number 458					
						3. Page 1 OF 3					
4. Title Ultimate Heat Sink Inoperable due to Boundary Valve Leakage											
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
03	24	2023	2023	- 001 -	01	08	17	2023	N/A		N/A
									Facility Name	☐ 052	Docket Number
									N/A		N/A
9. Operating Mode 4						10. Power Level 0%					
11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)											
10 CFR Part 20		☐ 20.2203(a)(2)(vi)		10 CFR Part 50		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		☐ 73.1200(a)	
☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		☐ 73.1200(b)	
☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		☐ 73.1200(c)	
☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		☐ 73.1200(d)	
☐ 20.2203(a)(2)(i)		10 CFR Part 20		☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(A)		10 CFR Part 73		☐ 73.1200(e)	
☐ 20.2203(a)(2)(ii)		☐ 21.2(c)		☐ 50.69(g)		☒ 50.73(a)(2)(v)(B)		☐ 73.77(a)(1)		☐ 73.1200(f)	
☐ 20.2203(a)(2)(iii)				☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(2)(i)		☐ 73.1200(g)	
☐ 20.2203(a)(2)(iv)				☒ 50.73(a)(2)(i)(B)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)		☐ 73.1200(h)	
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(C)		☐ 50.73(a)(2)(vii)					
☐ OTHER (Specify here, in abstract, or NRC 366A).											
12. Licensee Contact for this LER											
Licensee Contact Randy Crawford									Phone Number (include area code) 225-381-4177		
13. Complete One Line for each Component Failure Described in this Report											
Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS		
X	BS	ISV	J010	Y	X	BS	ISV	J010	Y		
14. Supplemental Report Expected									15. Expected Submission Date		
☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)									Month	Day	Year
16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced type written lines)											
On March 24, 2023, River Bend Station was in Mode 4 during Refuel Outage 22 when testing revealed leakage rates from Division 1 of the Standby Service Water system that could have previously rendered the Ultimate Heat Sink inoperable. On May 6, 2023, testing on Division 2 of the Standby Service Water system also resulted in leakage rates that could have previously rendered the Ultimate Heat Sink inoperable. These discoveries were made when River Bend Station was not in a required Mode of Applicability for the Ultimate Heat Sink. Corrective actions to restore compliance with the allowed design basis leakage were completed for both of the conditions described above prior to the completion of the refuel outage. These events are being reported in accordance with 10 CFR 50.73(a)(2)(v)(B) as an event that could have prevented fulfillment of the safety function provided by the Ultimate Heat Sink and 10 CFR 50.73(a)(2)(i)(B) as an operation or condition prohibited by Technical Specifications. The events are being reported in a single Licensee Event Report per guidance contained in NUREG-1022, Revision 3.											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to InfoCollection.Resource@nrc.gov, and the OMB reviewer at OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: OIRA-Comment@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME River Bend Station, Unit 1	☒ 050	2. DOCKET NUMBER 458	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 002 -	REV NO. - 01 -

NARRATIVE**EVENT DESCRIPTION**

On March 24, 2023, River Bend Station (RBS) was in Mode 4 during Refuel Outage 22 (RF22) when testing revealed leakage from the Standby Service Water (SSW) [BI] system that could have previously rendered the Ultimate Heat Sink (UHS) [BS] in-operable. The condition was discovered when a seat leakage test of SWP-MOV57A [ISV], Service Water Normal Supply Header A Inlet Isolation Valve resulted in a leak rate of approximately 30 gallons per minute (gpm). On May 6, 2023, while RBS was still in Mode 4 during RF22, a seat leakage test performed on SWP-MOV57B [ISV], Service Water Normal Supply Header B Inlet Isolation Valve resulted in a leak rate of 15.5 gpm. Design Basis calculations only allow a leakage rate of 6.9 gpm.

SSW and the UHS are required by Technical Specification (TS) 3.7.1 to be operable in Modes 1, 2, and 3. The UHS consists of one 200% cooling tower and one 100% capacity water storage basin. The UHS basin capacity is required by Reg Guide 1.27 and Updated Safety Analysis Report (USAR) 9.2.5 to maintain a minimum 30-day inventory to mitigate the consequences of a Design Basis Accident (DBA) without replenishment.

SWP-MOV57A was repaired by replacing the valve seat seals and adjusting the closing torque. SWP-MOV57B was repaired by adjusting the closing torque. Satisfactory post maintenance testing was completed for both valves.

This event is being reported in accordance with 10 CFR 50.73(a)(2)(v)(B) as an event that could have prevented fulfillment of the safety function provided by the Ultimate Heat Sink because the amount of leakage through SWP-MOV57A and SWP-MOV57B would not allow the UHS to fulfill the required 30-day mission time. The event is also being reported in accordance with 10 CFR 50.73(a)(2)(i)(B) as an operation or condition prohibited by Technical Specifications based on the past-operability review. Although this was a first-time performance of the SWP-MOV57A and SWP-MOV57B seat leakage test, the assumption is that the leakage existed prior to RF22 and TS 3.7.1 would have been applicable.

EVENT CAUSE

The direct cause of this event was excessive seat leakage past SWP-MOV57A and SWP-MOV57B.

The Root Cause of this event is that RBS did not challenge an assumption regarding the applicability of double isolation on the supply header of SSW. Because of these assumptions, SWP-MOV57A and SWP-MOV57B were not properly characterized by the In-Service Testing program and credit was taken for the isolation capability of check valves in-line with SWP-MOV57A and SWP-MOV57B.

SAFETY ASSESSMENT

A risk assessment was completed using assumptions of the worst-case ambient temperature and minimum initial water level in the UHS. The assessment concluded that although the UHS would not require replenishment until at least 22 days into an event, there is negligible severe accident risk associated with total leakage of 45.5 gpm (both trains combined) past SWP-MOV57A and SWP-MOV57B since the UHS would require replenishment prior to the 30-day mission time. Additionally, risk would be further reduced by consideration of actual annual temperature cycles which would significantly reduce evaporative loss from the UHS as well as proceduralized replenishment capabilities.

There were no actual Nuclear or Radiological safety consequences due to this event since the DBA did not occur. This event was of minimal significance to the health and safety of the public.



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NARRATIVE

CORRECTIVE ACTIONS

Completed:

1. Replace valve seat seals on SWP-MOV57A
2. Adjust closing torque on SWP-MOV57A and SWP-MOV57B
3. Root Cause Evaluation

Planned:

1. Re-categorize SWP-MOV57A and SWP-MOV57B to Category A in the IST basis documents.
2. Establish surveillance test procedures for leak testing SWP-MOV57A and SWP-MOV57B per IST basis documents.
3. Establish preventative maintenance tasks for internal inspection of SWP-MOV57A and SWP-MOV57B.
4. Issue Engineering Analysis to ensure an engineering basis document exists that clearly defines the Standby Service Water System boundaries and valve classifications, as well as actions required to maintain inventory.

PREVIOUS SIMILAR OCCURRENCES

LER 2014-004-00 was submitted by RBS on September 26, 2014, due to the UHS water inventory being less than the accident analysis required amount. The condition was a result of a design basis deficiency that did not account for the adverse effects of aggregate system leakage on compliance with the 30-day inventory required by Regulatory Guide 1.27.

Energy Industry Identification System (EIS) codes are identified in the text as [XX]. River Bend equipment codes are identified as (XX).